Predicting the Next Category and Region of a Visit

Speaker cards: 12 slides, about 15 minutes of talk, about 15 minutes for questions

Slot: 30 minutes *including* questions. First talk of the symposium, right after the opening (8:15). Audience: wireless networking and mobility management, not machine learning. Deck: `slides/main.pdf` (slides 1–11 numbered; backups B1–B10 after the last slide).

Time marks	S2 ends ≈ 2:30 · S5 ends ≈ 7:20 · S7 ends ≈ 10:30 · S9 ends ≈ 12:15 · end ≈ 15:15
If short of time	cut words in S3 and S6, never in S2 or S7
If ahead	add up to +60 s in S4 (the concatenation control), +40 s in S7

S1 · Title — 20 s

On screen: title, three authors, NESPeD-LAB / UFV.

OPEN "Good morning. I am Vitor Hugo, from the Federal University of Viçosa, in Brazil. This is joint work with Germano dos Santos and Fabrício Silva."

SAY

- One sentence and move on: "We ask whether one model can predict both what kind of place a person visits next and which neighborhood."

S2 · One person's day, and the question — 130 s

On screen: a schematic map, nine check-ins (coffee, gym, office, lunch), a "?", two boxes.

Rehearse this one aloud. It wins or loses the room at 8:30.

SAY

- "Here is one person's day: nine check-ins. Coffee, gym, office, lunch. The tenth is the question."
- "But nobody needs the tenth *place*. Texas alone has about a hundred and sixty thousand of them. A service needs two coarser things. What **kind** of place comes next: food, shopping, nightlife. And which **neighborhood**: a census tract. About a thousand of them in Alabama, eight and a half thousand in California."
- "If you know the neighborhood before the person gets there, you can push content to the cache that serves it, and plan capacity where demand is about to land. You already do this one level down, at the radio cell, with handover prediction. We are at the neighborhood. **A tract is not a cell, and we claim nothing about cells.**"
- "How far ahead? In our data the median gap from the last visit to the next runs from under half an hour in Florida to five and a half hours in Istanbul."
- "The obvious engineering answer is two models, one per question: two things to train, version and serve. One model for both is the shortcut, and it is known to fail: parameters shared between two jobs can settle on a compromise that is worse at both. So: **can one model answer both, and what does it cost?**"
- "The answer, so you can decide whether to keep listening: yes. One model, one forward pass. In California, ten neighborhoods out of 8,501 contain the right one 64 percent of the time. On the neighborhood question it outperforms a dedicated model at the two biggest datasets and is at worst 0.9 points behind at the others. And on the type-of-place question, how we represent a visit moved the score more than anything about the model. Here is how."

IF ASKED "Is that a real user?" No. It is a schematic of the input format, not a case from the data.

NEVER handover or cell-level gains as a claim. It is an analogy of scale only.

S3 · Check2HGI: a fourth level below the place — 100 s

On screen: the Check2HGI diagram, two lines below it.

OPEN "Most work gives each place one fixed vector. But the same coffee shop on a Tuesday morning and on a Saturday night is not the same signal."

SAY

- "So we add a fourth level below place, region and city: the check-in itself."
- "Each visit gets its own 64-dimensional vector, built from its place's category, the hour and day of week, and how long since the previous visit."
- "The graph is trained without labels: it never sees the next category or the next region."
- **Keep:** "It is built from the past alone. Edges between consecutive visits run forward in time only." (This is limit 3 and the answer to every leak question.)

NEVER "pre-trained encoder" for the per-visit input. Nothing pre-trained is on that path.

S4 · Result 1: the representation 115 s

On screen: the six-row table, check-in level vs place level.

OPEN "First result, and the most controlled one: only the input changes. Same model, same folds, same windows."

SAY

- "With a vector per visit instead of per place, next-category prediction improves at all six datasets and in every fold."
- "The gain is larger at the three smaller datasets, and under one point at the three largest. At Florida the fold-to-fold spread is larger than the gap, and the paired test does not separate the two."
- **Say this before anyone asks:** "And to be clear: most of that gain is already there if you just hand the place vector the hour and the category. What the graph adds beyond that, this control does not separate." (Backup B9.)

NUMBERS gain +0.23 to +2.81 macro-F1 · AL +1.62, AZ +2.58, Istanbul +2.81 · Florida +0.23, fold sd 0.42, $p = 0.067$.

IF ASKED CA and TX place cells: do not defend them as verified pairs. Their matching could not be checked from what is on disk. Both gains are under one point, so the message does not change.

S5 · Why sharing is not free, and the model 75 s

On screen: the model figure; "The risk", "The answer".

OPEN "Two questions, one model. Why is that not obviously a good idea?"

SAY

- "Because the parameters in the middle have to serve both jobs at once, and they can settle on a compromise that is worse for each than a dedicated model would have been. You get one artifact to deploy, and you may pay for it on both answers."
- *[point at the figure]* "So the model is built to limit that. The two inputs go into a shared trunk where each task can read the other's context. That is the part we want."
- "But the region task also keeps a **private** spatial path that the category task never touches, because where someone goes and what kind of place it is are not the same question. **The private path is the architecture admitting the risk.**"
- "Whether the shared part earns its keep is what we measured, and I will show you in two slides."

NEVER define "multi-task learning" as a concept; count blocks or widths. This is minute 7.

S6 · How it was measured 90 s

On screen: five rows: Regions, Split, Scored, Repeated, Decided.

OPEN "Before the verdict, how it was measured."

SAY

- "Regions are census tracts, from 520 in Istanbul to 8,501 in California."
- "We split by user: the visits of a test user never appear in training. Every number is on users the model has never seen."
- "Category is scored with macro-F1. Region with Acc@10: is the true region among the model's top ten?"
- "Twenty trained models per configuration: four seeds times five folds."

- **Never cut:** "And we wrote down what would count as 'not worse', two points, before we saw any result."

S7 · One model, two answers: the price 100 s

On screen: Dedicated vs Joint, six datasets, both tasks; green = outperforms.

OPEN "This is the price of one model instead of two."

SAY

- "On region, the one model is better at the two largest datasets, Texas and California, by about one point. Everywhere else it is behind by at most 0.87 points, inside the two-point margin we fixed before we saw any result."
- "Those four are real deficits, not ties. Every interval is entirely below zero."
- "On category it is ahead at Florida. The other five are unresolved, all within a fifth of a point of zero."
- **The thesis sentence:** "The trade is a measured one and not a free substitution."
- Say with the numbers, not as a confession: "The joint model is also the larger model, and this design does not separate size from sharing. And the two region wins sitting at the two biggest region counts is an observation, not a law: California has more regions than Texas and a smaller gain."

NUMBERS region: Texas +1.21, California +1.06 (20 of 20 folds); worst deficit Alabama −0.87 · category: Florida +0.19 (19 of 20 folds).

IF ASKED "Your Δ is not the difference of the columns." Right: Δ is the paired mean over the four seeds, so it can differ from the rounded columns in the second decimal.

NEVER "ties", "matches", "no difference", "the same" for the four region cells.

S8 · And against published systems 45 s

On screen: externals vs Dedicated vs Joint, both tasks.

SAY

- "Both our models, the dedicated and the joint, are above every system we compared against. The joint model by at least 4.11 points on category and 3.55 on region."
- "That margin belongs mostly to the representation from Result 1, so it is not evidence for sharing." Then move on.

IF ASKED the HMT-GRN column: it runs on our windows and folds, at one seed, and is read at its best validation epoch, the same rule as our models. STAN and ReHDM use their own splits.

NEVER "both our models by at least four points". The four-point margin is the joint model's only.

S9 · What a service would receive 60 s

On screen: lead time, unseen users, a shortlist; caveat box.

SAY

- "Lead time: the median gap to the next visit is from 0.4 hours in Florida to 5.5 hours in Istanbul, and 5 to 27 percent of the targets come more than three days later."
- "Unseen users: every number is on users whose visits never appeared in training."
- "A shortlist: ten regions out of 8,501 contain the true next one 64.54 percent of the time in California, and ten out of 6,553 do so 66.15 percent of the time in Texas."
- "We built and evaluated no such service. This is motivation, not a measured service result."

NEVER any distance in kilometers. See the Q&A card.

S10 · What it costs, and four limits 90 s

On screen: parameters (left), four limits (right).

SAY

- "The joint model is larger than the two dedicated models combined: 4.2 million against 1.9 million at Alabama, 5.2 against 2.8 at California. What one model gives is operational: one artifact to train,

version and deploy, one pass, two answers."
- Limit 1: "The representation is trained once over all users. Rebuilding it per fold from training users only moved the dedicated models by at most 0.17 macro-F1 and 0.13 Acc@10."
- Limit 2: "Epoch selection reads the fold the score is read on, so absolute scores are optimistic. The joint-versus-dedicated comparison is affected far less."
- Limit 3: "Each visit reads only the visits before it."
- Limit 4: "The joint model is larger, so its region result may come from size rather than sharing."

IF ASKED latency or energy: "About five million parameters and a window of nine visits per query. We did not measure latency or energy."

S11 · Conclusion 60 s

On screen: three bullets and one line.

SAY

- "The evidence does not attribute the outcome to sharing alone. On category, the representation moves the score more than the choice between one model and two."
- "One saved model reads both answers: above the dedicated model on region at Texas and California, within the two-point margin elsewhere."
- "These results do not mean that multi-task learning helps automatically."

OPEN Close with: "What a user will do next, and where, in a single forward pass instead of two. Thank you."

S12 · Thank you 30 s

On screen: Questions? Code, data, funders.

SAY

- "The code, the model, the representation, the baselines and the statistical tests are public at this link, and both datasets are public. This work was supported by FAPEMIG, CNPq and CAPES."
- **Leave this slide up during the questions.** Jump to a backup only when a question needs it, then come back.

Questions: the prepared answers

Question	Answer	Slide
"Your numbers do not match the paper." *(most likely: the PDF is in the proceedings)*	"The paper mixed sources from different versions; the corrected values come from the delivered build." For the HMT-GRN column: "the paper read HMT-GRN at a fixed final epoch; here it is read at its best validation epoch, the same rule as our models." That correction *favors* the competitor. Example if asked: "the Istanbul place-level cell was computed on an older build; matched on the same rows the gain is 2.81." The message survives: gain at all six datasets and every fold. Do not apologize or dramatize.	S8
"How do you know a visit vector does not see its target?"	By the mechanism: each visit node draws only on the visits before it, in training and at readout, and the graph is trained without the next-category or next-region label.	B4
"Do the external systems run on your folds?"	"Only HMT-GRN runs on our windows and folds. STAN and ReHDM use their own splits."	B7
"Where does the 2-point margin come from? Chosen after looking?"	Written in the analysis plan before any result was read, for region only. Category has no registered equivalence margin, so its differences are "unresolved", never a match. The two region gains are secondary results, outside the plan.	B2
"Would a Markov chain do this?"	A first-order Markov chain over regions gets 51.23 to 72.47 Acc@10. The joint model exceeds it by 4.1 to 10.0 at every dataset. Say the narrowest first: Florida, +4.07. Talk only about *our* margin over the floor.	B3
"Isn't this just the timestamp added to the place vector?" *(the most dangerous one)*	Largely, yes, on category: a control that adds the raw features to the place vector gains about as much (+1.73 / +1.70 / +1.02 at AL / AZ / FL). What the graph adds beyond that, this control does not separate.	B9
"Then what is multi-task learning for?"	On category the representation moves the score more than the choice between one model and two. Whether the exchange between tasks adds anything beyond it is not separated here. What one model gives for certain is operational.	B6
"What does one model instead of two cost?"	4.2 M vs 1.9 M parameters (AL), 5.2 M vs 2.8 M (CA), against the two dedicated models combined.	B5
"Does the gain grow with the number of regions?"	No. California has more regions than Texas and a smaller gain. An observation, not a law.	B8
"And the confidence intervals?"	The twelve cells with 90% intervals.	B10
"Why census tracts and not a grid?"	A grid cell is the usual target; we use official neighborhood-scale units, 520 to 8,501 per dataset.	B1
"How far off are the misses, in kilometers?" *(rehearse: it comes as a friendly question)*	"Not measured. It needs per-visit predictions the evaluation path does not keep. It is the first thing we would measure for a service." **Never give a distance.**	none
"Latency? Energy? Can it run on the edge?"	"About five million parameters and a window of nine visits per query. We did not measure latency or energy."	none
"You split by user, not by time. Drift? Retraining?"	"The split is by user and not by time, so the evaluation does not measure drift or the effect of sporadic events." The upside: every number is on unseen users.	none
"Gowalla is from 2009–2011."	Istanbul comes from Massive-STEPS, a recent collection, and Gowalla is the standard benchmark of this line. What matters is the pattern that repeats across continents and periods.	none

Question	Answer	Slide
"Privacy: the representation was trained on everyone."	Server side, on the operator's own logs; no privacy mechanism is claimed. Limit 1 measures the effect: rebuilding it with training users only moved the dedicated models by at most 0.17 macro-F1 and 0.13 Acc@10.	S10
"Only four runs?"	Each seed is a full five-fold cross-validation: twenty trained models per configuration. The paired difference varies 0.02 to 0.16 points across seeds.	S6
"Who would consume this?"	Name the consumer: load and demand anticipation at neighborhood level, caching, capacity planning, by an operator or a city service.	S9

Never say

- "beats" or "wins". The verb is **outperforms**.
- "ties", "matches", "no difference" for the four region cells. They are deficits inside a registered margin.
- Any distance in kilometers.
- Handover or cell-level gains as a claim. The tract is a neighborhood, not a radio cell.
- "Both our models by at least four points". The margins on S8 belong to the joint model.
- "The Markov floor beats the external systems". At its best epoch HMT-GRN is above the floor everywhere.
- "The gain comes from sharing". The design does not separate size from sharing.
- Any number read from the printed paper where the deck shows a different one. Use the slide.

On the day

- 8:30 in Paris is 4:30 in Brasília (France leaves summer time on 25 Oct).
- Arrive before the 8:15 opening and test the projector with the PDF.
- Carry the PDF on a USB stick and in the cloud. The deck is 16:9.
- The program publishes no room or chair. Introduce yourself to the chair before the session.
- Leave slide 12 (code, data, funders) on screen during questions.